9.7 Law of Sines and Law of Cosines

Name _____ Period ____ Date _____

Geometry 9.7 · Student sheet

I can…

- find the area of a triangle from two sides and the angle between them.
- use the Law of Sines to solve a triangle (AAS, ASA, or SSA).
- use the Law of Cosines to solve a triangle (SAS or SSS).

Vocabulary · fill in the missing word

Law of Sines — in any _____, $\sin\frac{A}{a} = \sin\frac{B}{b} = \sin\frac{C}{c}$

Law of Cosines — in any _____, $a^2 = b^2 + c^2 - 2bc\cos A$

Area of a triangle — Area = _____, using two sides and the angle between them

Part 1 · Discovery: does a ratio stay the same?

Draw any triangle that is **not** a right triangle, with sides at least 6 cm. Label A, B, C; side a is across from ∠A, and so on. Measure everything.

	Angle	Side across from it	Side ÷ sin(angle)
A		a =	
B		b =	
C		c =	

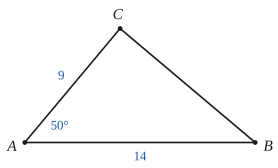
What do you notice about the last column?

Part 2 · Notes and examples (fill in each note, then work the example)

Area of a triangle: Area = ½ · b · c · _____ (two sides and the angle between them)

Example 1

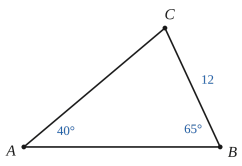
Find the area of △ABC with b = 9, c = 14, and m∠A = 50°. Round to the nearest tenth.



Law of Sines: sin A/a = _____ = _____. Use it for AAS, ASA, or SSA.

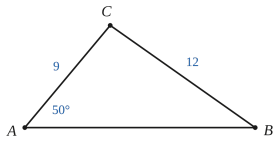
Example 2

Solve △ABC with m∠A = 40°, m∠B = 65°, and a = 12. Round to the nearest tenth.



Example 3

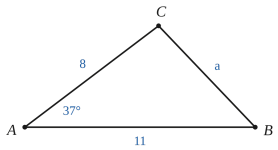
Solve △ABC with m∠A = 50°, a = 12, and b = 9. Round to the nearest tenth.



Law of Cosines: $a^2 = b^2 + c^2 -$ _____. Use it for SAS or SSS.

Example 4

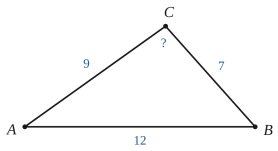
Find a in △ABC with b = 8, c = 11, and m∠A = 37°. Round to the nearest tenth.



SSS: find the _____ angle first (the one across from the longest side).

Example 5

A triangle has sides a = 7, b = 9, and c = 12. Find its largest angle to the nearest tenth of a degree.



Part 3 · Partner practice

Partner 1

Find the area of △ABC with a = 6, b = 10, and m∠C = 120°. Round to the nearest tenth.

Partner 2

Find c in △ABC with a = 5, b = 7, and m∠C = 60°. Round to the nearest tenth.